
ALGEBRA-SUMMER

Vincent Ferrara

Contents

0.1	Intro	3
1	Groups	5
1.1	Intro	5
1.1.1	Exercises	5
1.2	Dihedral Groups	6
1.2.1	Symmetry Group for an Equilateral Triangle	6
1.2.2	Symmetry Group for a Square	8
1.2.3	Symmetry Group of regular n-gon	9
1.2.4	Exercises	10
1.3	Symmetric Groups	11
1.3.1	Cycle Notation	12
1.3.2	Exercises	15
1.4	General Results on Groups	17
1.4.1	Subgroup Lattice	18
1.4.2	Exercises	21
1.5	Cyclic Groups	22

0.1 Intro

Theorem 0.1.1 (Division Algoritm). Suppose $a, b \in \mathbb{Z}$ s.t. $b > 0$. Then there are unique $q, r \in \mathbb{Z}$ s.t. $a = bq + r$ and $0 \leq r < b$

Proof. Let $a, b \in \mathbb{Z}$ s.t. $b > 0$. Consider $A = \{a - bx \mid x \in \mathbb{Z}\} \cap \mathbb{N}$

By the Well Ordering Principle since A has all non-negative terms it has a minimum lets call this r and call the associated $x \in \mathbb{Z}$, q .

Thus, $r = a - bq \implies a = bq + r$.

For contradiction, assume $r \geq b$.

$\implies r = b + m$ s.t. $m \geq 0$

$\implies b + m = a - bq \implies m = a - b(q + 1) \in A \implies m < r = \text{minimum} \implies \Leftarrow$ Thus $0 \leq r < b$

Fix $q_1, q_2, r_1, r_2 \in \mathbb{Z}$ s.t. $a = bq_1 + r_1 = bq_2 + r_2$ and $0 \leq r_1, r_2 < b$. WLOG let $r_1 \geq r_2$

$\implies bq_1 + r_1 = bq_2 + r_2 \implies r_1 - r_2 = b(q_2 - q_1) \implies r_1 - r_2$ is a multiple of b

Since $0 \leq r_1, r_2 < b \implies 0 \leq r_1 - r_2 < b$

Since $r_1 - r_2$ is strictly less than b the only multiple of b it could be is 0 thus $r_1 = r_2$

$\implies q_1 = q_2$ □

Lemma 0.1.2. (Bezout's Lemma) Suppose $a, b \in \mathbb{Z} \setminus \{0\}$. Then there are $x, y \in \mathbb{Z}$ s.t. $ax + by = \gcd(a, b)$.

Proof. Let $a, b \in \mathbb{Z} \setminus \{0\}$. Consider $A = \{ax + by \mid x, y \in \mathbb{Z}\} \cap \mathbb{N}$

By the Well Ordering Principle since A has all non-negative terms it has a minimum lets call this d and call the associated $x, y \in \mathbb{Z}$, u and v .

By Theorem 0.1.1 $\exists q, r \in \mathbb{Z}$ s.t. $a = dq + r$ and $0 \leq r < d$

Case 1: $r = 0 \implies d \mid a$

Case 2: $0 < r < d \implies r = a - dq \implies r = a - (au + bv)q = a(1 - uq) + b(-vq) \in A$

$\implies r < d = \text{minimum} \implies \Leftarrow$

Thus $d \mid a$ and similarly $d \mid b$

Let $c \in \mathbb{N}$ s.t. $c \mid a$ and $c \mid b$

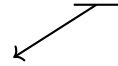
$\implies a = cm$ and $b = cn$ for $m, n \in \mathbb{Z}$

$d = au + bv = c(mu) + c(mv) = c(mu + mv) \implies c \mid d \implies d = \gcd(a, b)$ □

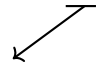
Algorithm 0.1.3. (Euclidean Algorithm)

Find $\gcd(272, 571)$ and $x, y \in \mathbb{Z}$ s.t. $272x + 571y = \gcd(272, 571)$

$$571 = 272 \cdot 2 + 27 \quad (1)$$



$$272 = 27 \cdot 10 + 2 \quad (2)$$



$$27 = 2 \cdot 13 + 1 \quad (3)$$

Last term before remainder is zero is gcd

$$\gcd(272, 571) = 1$$

$$2 = 272 - 27 \cdot 10$$

Use equation (2)

$$= 272(21) - 571(10)$$

$$1 = (571 - 272 \cdot 2) - 13(272(21) - 571)$$

Use equation (3) with (2)

$$= 571(131) + 272(-275)$$

1 Groups

1.1 Intro

Definition 1.1.1. A group is a set G together with a binary operation $*$: $G \times G \rightarrow G$ s.t.

- **(Closure)** For all $a, b \in G$ we have $a * b \in G$
- **(Identity)** There is $e \in G$ s.t. for all $a \in G$ $a * e = e * a = a$
- **(Inverses)** For all $a \in G$ there is an $a^{-1} \in G$ s.t. $a * a^{-1} = a^{-1} * a = e$
- **(Associativity)** For all $a, b, c \in G$ we have $a * (b * c) = (a * b) * c$

Definition 1.1.2. Given a group G and $g \in G$ we say the order of g is $n \in \mathbb{N}$ if n is the minimum value s.t. $g^n = g * g * \dots * g = e$
Write as: $\text{ord}(g) = |g| = n$

1.1.1 Exercises

Example 1.1.3. Let $G = \mathbb{R} \setminus \{-1\}$ and define $a * b = a + b + ab$. Show $(G, *)$ is an Abelian group

- (Abelian) For all $a, b \in G$ $a * b = a + b + ab = b + a + ba = b * a$
- (Closure) $\checkmark$
- (Identity) For all $a \in G$ $a * 0 = 0 * a = 0$
- (Inverses) For all $a \in G$ let $b = \frac{-a}{a+1}$.

$$\text{Thus, } a * b = b * a = a + b + ab = a + b(a+1) = \frac{-a}{a+1}(a+1) = a - a = 0$$

- (Associativity) For all $a, b, c \in G$

$$a * (b * c) = a * (b + c + bc) = a + b + c + bc + ab + ac + abc = (a + b + ab) * c = (a * b) * c$$

Therefore, G is an Abelian group

Example 1.1.4. Find all $x \in \mathbb{Z}$ s.t. $5x \equiv 4 \pmod{9}$

$$5x \equiv 4 \pmod{9}$$

$$x \equiv 8 \pmod{9}$$

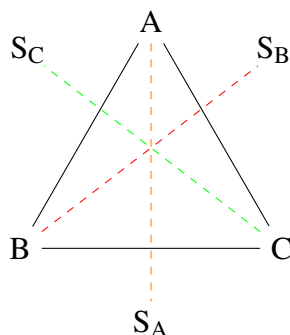
$$x = 9n + 8$$

$$2 * 5 = 10 \equiv 1 \pmod{9}$$

$$n \in \mathbb{Z}$$

1.2 Dihedral Groups

1.2.1 Rigid Motions of an Equilateral Triangle



Rotations

$r = 120^\circ$ rotation

$r^2 = 240^\circ$ rotation

$r^3 = 360^\circ$ rotation $= e = 0^\circ$ rotation

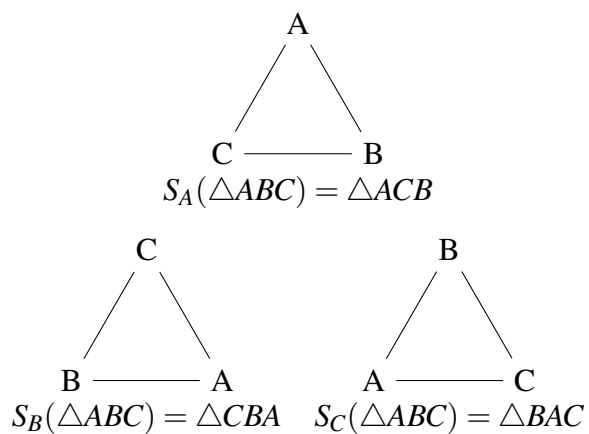
Reflections

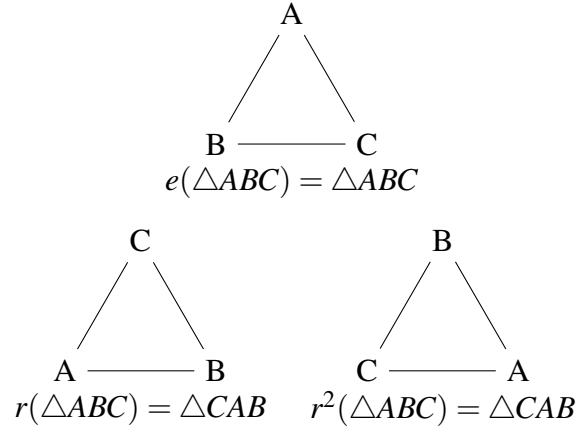
$S_A : \triangle ABC \rightarrow \triangle ACB$

$S_B : \triangle ABC \rightarrow \triangle CBA$

$S_C : \triangle ABC \rightarrow \triangle BAC$

Transformation Diagrams





Operations

$$\begin{aligned}
 r \circ S_A(\triangle ABC) &= r(\triangle ACB) = \triangle BAC = S_C(\triangle ABC) & r \circ S_A &= S_C \\
 r^2 \circ S_A(\triangle ABC) &= r^2(\triangle ACB) = \triangle CBA = S_B(\triangle ABC) & r^2 \circ S_A &= S_B
 \end{aligned}$$

Now we only need S_A so lets call this S

Thus we only need S, r, r^2, S, r, r^2, e

Also $S^2 = r^3 = e$

How S, r, r^2 Commute

$$\begin{aligned}
 r \circ S(\triangle ABC) &= r(\triangle ACB) = \triangle BAC = S \circ r^2 \\
 r^2 \circ S(\triangle ABC) &= r^2(\triangle ACB) = \triangle CBA = S \circ r
 \end{aligned}$$

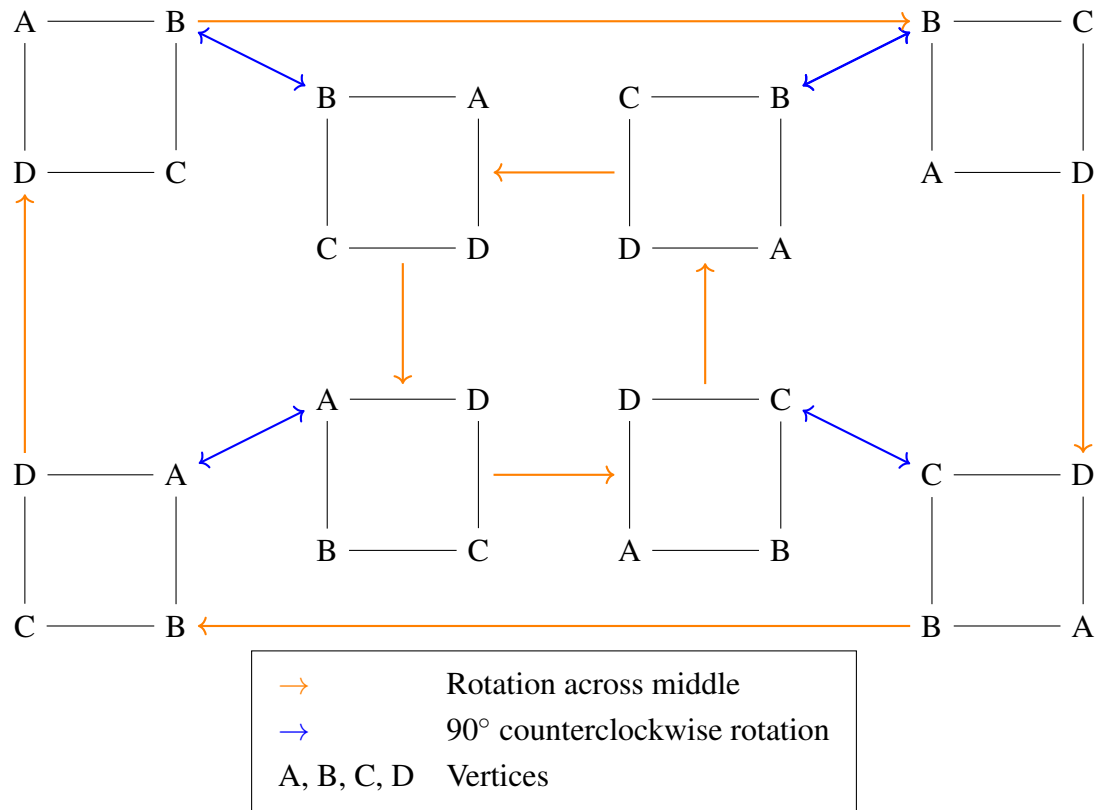
Definition 1.2.1. The group of symmetries of an equilateral triangle is $D_3 = \{e, r, r^2, S, Sr, Sr^2\}$, where $r^3 = S^2 = e$, $rS = Sr^2$

Dihedral group of order $2 \cdot 3 = 6$

Compact Notation $D_3 = \langle r, s^1 \mid r^3 = S^2 = e, rS = Sr^2 \rangle$

¹The generators of D_3 meaning you can create all other operations from them

1.2.2 Symmetry Group for a Square



Rotations

$r = 90^\circ$ rotation

$r^2 = 180^\circ$ rotation

$r^3 = 270^\circ$ rotation

$r^4 = 360^\circ$ rotation = $e = 0^\circ$ rotation

Reflections

$$s : \square ABCD \rightarrow \square BCDA$$

How does s, r Commute

From the diagram we see that $sr = r^3s \implies rs = sr^3$

Definition 1.2.2. The group of symmetries of a square is $D_4 = \{e, r, r^2, r^3, s, sr, sr^2, sr^3\}$

Dihedral group of order $2 \cdot 4 = 8$

$D_4 = \langle s, r \mid r^4 = s^2 = e, sr = r^3s \rangle$

1.2.3 Symmetry Group of regular n-gon

Rotation: $r = \frac{360^\circ}{n} \circlearrowleft (e, r, r^2, \dots, r^{n-1})$

Reflection: $s = \text{reflection across middle } (s, sr, sr^2, \dots, sr^{n-1})$

Claim: $rs = sr^{n-1}$.

Proof. We only need 3 points since our shape is defined by the order of those 3 points

$$r \circ s(n, 1, 2) = r(2, 1, n) = (1, n, n-1)$$

$$s \circ r^{n-1}(n, 1, 2) = s(n-1, n, 1) = (1, n, n-1)$$

$$\text{Thus } rs = sr^{n-1}$$

□

Definition 1.2.3. The dihedral group D_n is the symmetry group of a regular n-gon.

$$D_n = \langle s, r \mid r^n = s^2 = e, rs = sr^{n-1} \rangle$$

Claim: For all $1 \leq m \leq n-1$ we have $r^m s = sr^{n-m}$.

Proof. Base Case: $m = 1 \implies rs = sr^{n-1} \checkmark$

Inductive Step: Assume for $k \geq 1$ the statement is true

Consider:

$$r^{k+1}s = rr^k s$$

$$= rsr^{n-k}$$

$$= sr^{n-1}r^{n-k}$$

$$= sr^n r^{n-(k+1)}$$

$$= sr^{n-(k+1)}$$

Apply inductive hypothesis on $r^k s$

□

1.2.4 Exercises

Example 1.2.4. Simplfy

- $(sr^5)r(sr^3)$ in D_4

$$= (sr)r(rs)$$

$$= e$$

- $(sr^4)(sr^2)$ in D_6

$$= (r^2s)(sr^2) \quad \text{Use } r^m s = sr^{n-m}$$

$$= r^4$$

- $r^3(sr)$ in D_4

$$= r^3(r^3s)$$

$$= r^2s$$

Example 1.2.5. The Cayley table of the D_3 can be written

	e	r	r^2	s	rs	r^2s
e	e	r	r^2	s	rs	r^2s
r	r	r^2	e	rs	r^2s	s
r^2	r^2	e	r	r^2s	s	rs
s	s	r^2s	rs	e	r^2	r
rs	rs	s	r^2s	r	e	r^2
r^2s	r^2s	rs	s	r^2	r	e

1.3 Symmetric Groups

Definition 1.3.1. Given a set X , a permutation of X is a bijection $X \rightarrow X$
 $S_X = \{ \sigma : X \rightarrow X \mid \sigma \text{ is a bijection} \}$

Lemma 1.3.2. For all sets $X \neq \emptyset$, S_X is a group

Proof.

- Closure: Suppose $\sigma, \mu \in S_X$
 Consider $\sigma \circ \mu$ since it is a composition of bijections it is also a bijection.
 Also functions go from $X \rightarrow X$ thus $\sigma \circ \mu$ is the same.
 Therefore, the composition is a bijection that is mapped from $X \rightarrow X$ thus $\sigma \circ \mu \in S_X$
- Identity: Consider $e : X \rightarrow X$ s.t. $e(x) \mapsto x$
 Thus $e \in S_X$ and for all $\sigma \in S_X$ $\sigma \circ e(x) = \sigma(x)$
 Thus $\sigma \circ e = \sigma$. Also, it follows that $e \circ \sigma = \sigma$
- Inverses: Since S_X has only bijective functions all bijective functions have inverses.
- Associativity: We have associativity by the fact that function composition has associativity.

□

Definition 1.3.3. The permutation group on n -objects S_n is the S_X with $X = [n]$
 Observe $|S_n| = n!$

Example 1.3.4. $\sigma, \mu \in S_4$

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \mu = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix}$$

$$\sigma^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \mu^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix}$$

$$\sigma\mu = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \mu\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix}$$

1.3.1 Cycle Notation

Example 1.3.5. $\sigma, \mu \in S_5$

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 2 & 5 & 1 \end{pmatrix} \mu = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 2 & 1 & 3 \end{pmatrix}$$

$$\sigma = (13245) \mu = (14)(253)$$

Example Calculations:

$$\sigma\mu = (13245)(14)(253)$$

$$= (152)(34)$$

$$\mu\sigma = (14)(253)(13245)$$

$$= (12)(354)$$

$$\mu^2 = (14)(253)(14)(253)$$

$$= (14)^2(253)^2 = (235)$$

Lemma 1.3.6. *Disjoint cycles are commutative*

Proof. Let α, β be disjoint cycles of $[n]$

Let $\alpha = (a_1, a_2, \dots, a_k)$, and $\beta = (b_1, b_2, \dots, b_j)$

Let $x \in [n]$

Consider: $\alpha\beta(x), \beta\alpha(x)$

If $x \in \alpha$ then $x \notin \beta \implies \beta(x) = x \implies \alpha\beta(x) = \alpha(x) = a_i$

$\implies \beta\alpha(x) = \beta(a_i) = a_i$ since $a_i \notin \beta$

Use similar reasoning for if $x \in \beta$

$\implies \alpha\beta = \beta\alpha$

□

Lemma 1.3.7. *Every $\sigma \in S_n$ can be written as disjoint cycles*

Lemma 1.3.8. *The order of an m -cycle is m*

Lemma 1.3.9. Suppose G is a group and $\text{ord}(g) = n$. Then $g^m = e \iff n \mid m$

Proof. Let G be a group and $\text{ord}(g) = n$

($\Leftarrow$)

Assume $n \mid m$

$$\implies m = nk, k \in \mathbb{Z}$$

$$g^m = g^n k = (g^n)^k = (e)^k = e$$

($\Rightarrow$)

Assume $g^m = e$ and use the division algorithm (0.1.1) to write $m = nq + r$ s.t. $0 \leq r < n$

$$e = g^m = g^{nq+r} = (g^n)^q g^r = g^r \implies r = 0 \implies n \mid m$$

□

Theorem 1.3.10. If $\sigma_1, \sigma_2, \dots, \sigma_k$ are disjoint cycles of length $a_1, a_2, \dots, a_k$ and $\sigma = \sigma_1 \sigma_2 \dots \sigma_k$. Then $\text{ord}(\sigma) = \text{lcm}(a_1, a_2, \dots, a_k)$.

Proof. Let $m = \text{ord}(\sigma)$, $l = \text{lcm}(a_1, a_2, \dots, a_k)$

Since l is a lcm it is also a multiple so $l = a_j q_j$ for $1 \leq j \leq k$ where $q_j \in \mathbb{Z}$

$$\begin{aligned} \sigma^l &= \sigma_1^l \sigma_2^l \dots \sigma_k^l \\ &= (\sigma_1^{a_1})^{q_1} \dots (\sigma_k^{a_k})^{q_k} \\ &= (1) \end{aligned}$$

We can do this because of Lemma 1.3.6

Apply Lemma 1.3.8

$$\implies m \mid l$$

Also, $(1) = \sigma^m = \sigma_1^m \dots \sigma_k^m$ since they are disjoint none of them are inverses of each other

$$\implies \sigma_1^m = \dots = \sigma_k^m = (1)$$

$$(\text{Lemma 2.1.9}) \implies a_1 \mid m \dots a_k \mid m \implies l \mid m$$

Thus, since $l \mid m$ and $m \mid l$ $m = l$

□

Example 1.3.11.

- Find an element of order 12 in S_7
 $(1234)(567) \in S_7$ by Theorem 1.3.10 $\implies$ the order is $3 \cdot 4 = 12 = \text{lcm}(3,4)$
- Possible orders of elements in S_5
 - 1: (1) , $5=1+1+1+1+1$
 - 2: (12) , $5=2+1+1+1+1$ or $(12)(34)$, $5=2+2+1$
 - 3: (123) , $5=3+1+1$
 - 4: (1234) , $5=4+1$
 - 5: (12345) , $5=5$
 - 6: $(12)(345)$, $5=2+3$

1.3.2 Exercises

Example 1.3.12. Write the following as the product of disjoint cycles and find the order.

- $(1358)(378)(125) \in S_8$
 (128537) order=6
- $(1264)^{-1}(364)^2 \in S_6$
 $= (1462)(346)$
 $= (142)(36)$ order=6

Example 1.3.13. Find all Possible orders of elements in S_7

- 1: (1) , $7=1+1+1+1+1+1+1$
- 2: (12) , $7=2+1+1+1+1+1$ or $(12)(34)$, $7=2+2+1+1+1$ or $(12)(34)(56)$, $7=2+2+2+1$
- 3: (123) , $7=3+1+1+1+1$ or $(123)(456)$, $7=3+3+1$
- 4: (1234) , $7=4+1+1+1$ or $(1234)(56)$, $7=4+2+1+1$
- 5: (12345) , $7=5+1+1$
- 6: (123456) , $7=6+1$ or $(123)(45)$, $7=3+2+1+1$, or $(123)(45)(67)$, $7=3+2+2+1$
- 7: (1234567) , $7=7$
- 10: $(12345)(67)$, $7=5+2$
- 12: $(1234)(567)$, $7=4+3$

Example 1.3.14. What is the largest possible order of an element in S_{10} ?

cycle type	order
(5,1,1,1,1)	5
(5,2,1,1,1)	10
(5,2,2,1)	10
(5,3,1,1)	15
(5,3,2)	30
(5,4,1)	20
(5,5)	5
(6,1,1,1,1)	6
(6,2,1,1)	6
(6,3,1)	6
(6,4)	12
(7,1,1,1)	7
(7,2,1)	14
(7,3)	21
(8,1,1)	8
(8,2)	8
(9,1)	9
(10)	10

Max=30

1.4 General Results on Groups

Lemma 1.4.1. *Suppose G is a group. Then*

- (1) *The identity is unique*
- (2) *Inverses are unique*
- (3) *For every $a, b, c \in G$ there is a unique $x \in G$ s.t. $ax = b$*
- (4) *If $ab = ac$ then $b = c$, similarly if $xy = zy$ then $x = z$*
- (5) $(ab)^{-1} = b^{-1}a^{-1}$
- (6) *If $(ab)^2 = a^2b^2$ then $ab = ba$*
- (7) *If $\text{ord}(g) = 2$ for all $g \in G$ then G is abelian*

• *Proof.* (3) Let $ax = b \implies x = a^{-1}b$

Suppose $x, y \in G$ s.t. $ax = b$ and $ay = b$

$$ax = ay \implies x = y \quad \square$$

• *Proof.* (6) Assume $(ab)^2 = a^2b^2$

$$ab = (a^{-1}a)(ab)(bb^{-1}) = a^{-1}a^2b^2b^{-1} = a^{-1}(ab)^2b^{-1} = (a^{-1}a)(ba)(bb^{-1}) = ba \quad \square$$

• *Proof.* Assume $\text{ord}(g) = 2$ for all $g \in G$

$$\implies g = g^{-1} \text{ for all } g \in G$$

Let $a, b \in G$. Consider ab

$$ab = (ab)^{-1} = b^{-1}a^{-1} = ba \quad \square$$

Definition 1.4.2. Given a group G and a subset $H \subseteq G$, we say H is a subgroup $H \leq G$ if

- The identity of G is in H
- For all $h \in H, h^{-1} \in H$
- For all $a, b \in H, ab \in H$

Theorem 1.4.3. Suppose that G is a group and $\emptyset \neq H \subseteq G$
 then $H \leq G \iff$ for all $a, b \in H, ab^{-1} \in H$

Proof.

($\implies$) Assume $H \leq G$

Let $a, b \in H$

By Definition 1.4.2 $\implies b^{-1} \in H \implies ab^{-1} \in H$

($\impliedby$) Assume for all $a, b \in H, ab^{-1} \in H$

- (Identity) Consider $a \in H$

$$\implies aa^{-1} = e \in H$$

- (Inverses) $\checkmark$

- (Closure) Consider $a, b^{-1} \in H$

$$\implies ab \in H$$

□

1.4.1 Subgroup Lattice

Subgroup of Order 4

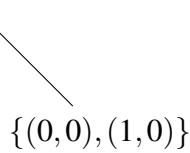
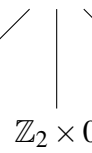
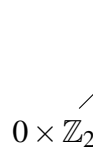
$$\mathbb{Z}_4 = \{0, 1, 2, 3\}$$



$$\begin{array}{c} | \\ 0 = \{0\} \end{array}$$

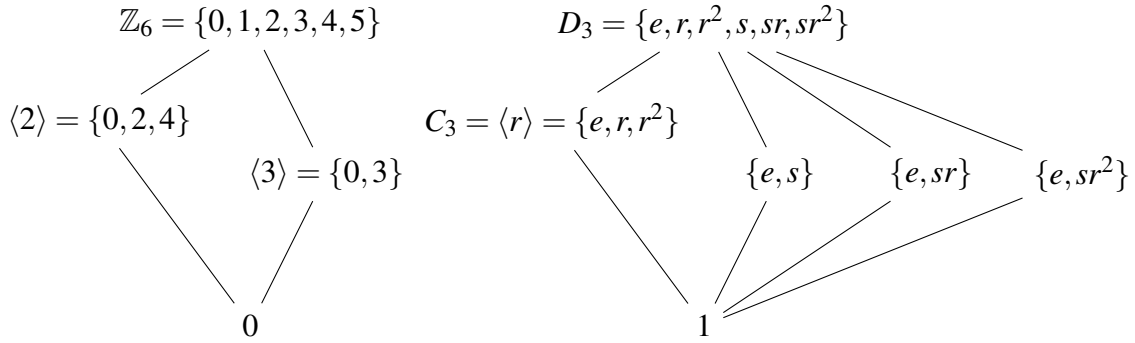
$$0 = \{0\}$$

$$\mathbb{Z}_2 \times \mathbb{Z}_2 = \{(0, 0), (1, 0), (0, 1), (1, 1)\}$$



$$0 = \{(0, 0)\}$$

Subgroup of Order 6



Definition 1.4.4. Given a group G and a subset $S \subseteq G$

- The centralizer of S in G : $C_G(S) = \{g \in G \mid gs = sg, \forall s \in S\}$
- The center of G : $(C_G(G)) = Z(G) = \{z \in G \mid gz = zg, \forall g \in G\}$

Lemma 1.4.5. $C_G(S)$ and $Z(G)$ are subgroups of G

Proof. Both contain e so they are nonempty

Let $a, b \in C_G(S)$ and $s \in S$

Consider:

$$\begin{aligned}
 as &= sa \\
 sab^{-1} &= asb^{-1} \\
 &= a(b^{-1}s)b \\
 &= (ab^{-1})s
 \end{aligned}$$

Thus $ab^{-1} \in C_G(S)$

Thus, by Theorem 1.4.3 $C_G(S)$ is a subgroup

$\implies Z(G) \leq G$ since the center is a special case of the centralizer

□

Example 1.4.6. Centralizers in D_4

- $C_{D_4}(e) = D_4 = C_{D_4}(r^2)$ ²
- $C_{D_4}(r) = e, r, r^2, r^3 = C_{D_4}(r^3)$
- $C_{D_4}(s) = e, s, r^2, sr^2 = C_{D_4}(sr^2)$
- $C_{D_4}(sr) = e, sr, r^2, sr^3 = C_{D_4}(sr^3)$
- $Z(D_4) = e, r^2$

²Since $r^2s = sr^2$ and $r^2r = rr^2$

1.4.2 Exercises

Example 1.4.7. Prove that if G is abelian if and only if for all $a, b \in G$ we have $(ab)^{-1} = a^{-1}b^{-1}$

Proof. ($\implies$) Assume G is abelian and let $a, b \in G$

$$\implies (ab)^{-1} = b^{-1}a^{-1} = a^{-1}b^{-1}$$

($\impliedby$) Assume for all $a, b \in G$ we have $(ab)^{-1} = a^{-1}b^{-1}$

Consider: a^{-1} and b^{-1}

$$ba = (a^{-1}b^{-1})^{-1} = ab$$

□

Example 1.4.8. Suppose G is a group of order 6 containing 7 and 19 where the operation is multiplication modulo 36. List all the elements of G .

- $7 \times 7 = 49 = 13$
- $19 \times 19 = 361 = 1$
- $7 \times 19 = 133 = 25$
- $13 \times 19 = 247 = 31$

$$G = \{1, 7, 13, 19, 25, 31\}$$

Example 1.4.9. Suppose that G is a group such that for all $a, b, c, d \in G$ such that if

$$axb = cxd$$

then $ab = cd$. Show that G is abelian

Proof. Consider: $a, b \in G$

$$aa^{-1}b = ba^{-1}a \implies ab = ba$$

□

1.5 Cyclic Groups

Definition 1.5.1. Given a group G and an element $g \in G$, the cyclic subgroup generated by g is $\langle g \rangle = \{g^m \mid m \in \mathbb{Z}\} = \{\dots, g^{-2}, g^{-1}, e, g, g^2, \dots\}$

We say G is a cyclic group if there is a $g \in G$ s.t. $G = \langle g \rangle$

$$\text{ord}(g) = |\langle g \rangle|$$

Cyclic subgroups of $\mathbb{Z}$

Example 1.5.2.

- $\langle 2 \rangle = \{2m \mid m \in \mathbb{Z}\} = \{\dots, -4, -2, 0, 2, 4, \dots\} = 2\mathbb{Z}$
- $\langle n \rangle = \{nm \mid m \in \mathbb{Z}\} = \{\dots, -2n, -n, 0, n, 2n, \dots\} = n\mathbb{Z}$

Cyclic subgroups of $\mathbb{Q}^\times$

Example 1.5.3.

- $\langle -1 \rangle = \{-1, 1\}$
- $\langle 2 \rangle = \{2^m \mid m \in \mathbb{Z}\}$

Cyclic subgroups of $\mathbb{C}^\times$

Example 1.5.4.

$$\langle i \rangle = \{i, -1, -i, 1\}$$

Cyclic subgroups of $\mathbb{Z}_{12}$

Example 1.5.5.

- $\mathbb{Z}_{12} = \langle 1 \rangle = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$
- $\langle 2 \rangle = \{0, 2, 4, 6, 8, 10\}$
- $\langle 4 \rangle = \{0, 4, 8\}$
- $\langle 3 \rangle = \{0, 3, 6, 9\}$

Cyclic subgroups of $U_9 = \{1, 2, 4, 5, 7, 8\}$

Example 1.5.6.

- $\langle 1 \rangle = \{1\}$
- $\langle 8 \rangle = \{8, 64\} = \{1, 8\}$
- $\langle 4 \rangle = \{4, 16, 64\} = \{1, 4, 7\}$
- $\langle 2 \rangle = \{2, 4, 8, 16, 32, 64\} = \{1, 2, 4, 5, 7, 8\} = U_9$

Cyclic subgroups of D_n

Example 1.5.7.

- $\langle r \rangle = \{e, r, r^2, \dots, r^{n-1}\} = Z_n^3 = C_n^4$
- $\langle s \rangle = \{e, s\}$
- $\langle sr \rangle = \{e, sr\}$

Cyclic subgroups of S_5

Example 1.5.8.

$$\langle (12)(345) \rangle^5 = \{(1), (12)(345), (354), (12), (345), (12)(354)\}$$

Theorem 1.5.9. Suppose $g \in G$ with $\text{ord}(g) = n$ and let $m \in \mathbb{N}$ then, $\langle g^m \rangle = \langle g^{\gcd(m,n)} \rangle$

Proof. Set $d = \gcd(m, n)$.

Take $x \in \langle g^m \rangle \implies x = (g^m)^k = g^{mk}$ for some $k \in \mathbb{Z}$

Also, $m = dq$ for some $q \in \mathbb{Z}$ since d is a divisor of m

$x = g^{dqk} = (g^d)^{qk} \in \langle g^d \rangle$ so $\langle g \rangle \subseteq \langle g^d \rangle$

By Lemma 0.1.2 there exists $r, s \in \mathbb{Z}$ s.t. $mr + ns = d$

Take $x \in \langle g^d \rangle \implies x = g^{dk}$ for some $k \in \mathbb{Z}$

$g^{dk} = g^{k(mr+ns)} = g^{mkr}(g^n)^{ks} = g^{mkr} \in \langle g^m \rangle$ so $\langle g^d \rangle \subseteq \langle g^m \rangle$

Thus $\langle g^m \rangle = \langle g^d \rangle$

□

³not $\mathbb{Z}$

⁴Multiplicative cyclic group of order n

⁵By Theorem 1.3.10 the $\text{lcm}(2,3)=6=\text{order}$ and thus this cyclic group will have 6 elements

Theorem 1.5.10 (Fundamental Theorem of Cyclic Groups).

- I. Every subgroup of a cyclic group is cyclic*
- II. If G is a cyclic group of order n and $d \mid n$ then there is a unique subgroup $H \leq G$ with $|H| = d$*

I. Proof. Suppose $G = \langle g \rangle$ and $H \leq G$

Case 1: $H = \{e\} = \langle e \rangle \checkmark$

Case 2: $H \neq \{e\}$. Thus, $H = \{\dots, g^{-m_2}, g^{-m_1}, e, g^{m_1}, g^{m_2}, \dots\}$ s.t. $m_i \in \mathbb{N}$

By the Well Ordering Principle, there exists a minimum m_i call this m

Since $g^m \in H \implies (g^m)^n \in H \ \forall n \in \mathbb{Z} \implies \langle g^m \rangle \subseteq H$

Take $h \in H$, write $h = g^k$ for $k \in \mathbb{Z}$

By the division algorithm (Thm. 0.1.1), there exists a $q, r \in \mathbb{Z}$ s.t.

$k = mq + r$ with $0 \leq r < m$

$g^r = (g^k)(g^{-mq})$ thus since $g^k \in H$ and $g^{-mq} \in \langle g^m \rangle \subseteq H \implies g^r \in H$

Since $0 \leq r < m$ and m is the minimum $\implies r = 0$

Thus $h = (g^m)^q \in \langle g^m \rangle \implies H \subseteq \langle g^m \rangle$

$\implies H = \langle g^m \rangle \implies H$ is cyclic □

II. Proof. Assume G is a cyclic group of order n and $d \mid n$

Let $G = \langle g \rangle$

Existence: Consider $\langle g^{\frac{n}{d}} \rangle$ this has d elements so $\checkmark$

Uniqueness: Let $H \leq G$, $|H| = d$ by Theorem 1.5.10.I

$\implies H = \langle g^m \rangle$ by Theorem 1.5.9 $\implies H = \langle g^{\gcd(m,n)} \rangle$

Thus $|H| = d = \frac{n}{\gcd(m,n)} \implies \gcd(m,n) = \frac{n}{d} \implies H = \langle g^{\frac{n}{d}} \rangle$ which unique describes a group □

Definition 1.5.11 (Euler's Totient Function).

For $n \in \mathbb{N}$, define $\varphi(n) = \left| \{1 \leq m \leq n \mid \gcd(m,n) = 1\} \right|$

Theorem 1.5.12.

Suppose G is a cyclic group of order n and $d \mid n$ then the number of elements of G of order d is $\varphi(d)$

Proof. Assume G is cyclic group of order n and $d \mid n$

Let $H \leq G$ be the unique subgroup s.t. $|H| = d \iff$ Theorem 1.5.10.II

Let $H = \langle h \rangle \iff$ Theorem 1.5.10.I

Let $x \in G$ have order d

Consider: $\langle x \rangle$ since x is order $d \implies |\langle x \rangle| = d$

Since H is unique $\implies \langle h \rangle = \langle x \rangle \implies x = h^m$ where $\gcd(m, d) = 1 \iff$ Theorem 1.5.9

The number of choices of m is $\varphi(d)$

□